Final Project Backlog — 2048

Advanced topics: Git/GitHub branching workflows and stack-based Undo.

#	Name	How to Demo	Notes
1	Display the game interface	Launch the application; the 4×4 board, numbered tiles, score display and controls are visible.	Java Swing GUI with custom rounded tiles and panels.
2	Generate starting tiles	Start a new game; two numbered tiles appear at random empty positions.	Initialize the board with two tiles.
3	Move tiles	Press the arrow keys; tiles slide in the selected direction.	Support movement left, right, up and down.
4	Merge matching tiles	Move equal-valued tiles together; they combine into a tile with double the value.	Each tile merges at most once per move.
5	Generate tiles after valid moves	Make a move that changes the board; a new tile appears in an empty cell.	Do not generate a tile after an ineffective move.
6	Track the score	Merge matching tiles; the displayed score increases.	Add the value of each resulting merged tile to the score.
7	Detect game over	Fill the board so no moves or merges remain; the game-over message appears.	Check both empty spaces and possible adjacent merges.
8	Detect reaching 2048	Create a 2048 tile; the victory message appears.	Detect when the winning tile is created.
9	Restart the game	Click New Game during play; the board and current score reset.	Start a fresh game without restarting the application.

10	Save high score	Achieve a high score, close the game and reopen it; the highest score remains displayed.	Persist the high score using Java Preferences.
11	Collaborate using GitHub	Open the repository and show the branch and commit history.	Use separate branches and merge completed work into `main`.
12	Undo previous moves	Make several valid moves, then click Undo repeatedly; earlier board states and scores return.	Use Java stacks to store board and score snapshots. No fixed Undo limit.
13	Provide on-screen movement controls	Click the four directional buttons; tiles move as they do with the arrow keys.	Provide mouse-based controls alongside keyboard input.
14	Display custom rounded tiles	Launch the game and observe the tile appearance as the board changes.	Use custom Swing painting for rounded tile backgrounds.
15	Build and run using Maven	Run `mvn clean package`, followed by the documented Maven launch command; the game opens.	Maven project configured for JDK 26.